

- Two-way protection in all services through certificate-based authentication like Public Key Infrastructure (PKI) and Organization based Access Control (OrBAC) for capability-based access control.

Azure IoT Hub and Digital Twins

All electronic devices in a smart home are connected in a controllable network. This allows users to interact with their homes through these connected devices from anywhere in the world. The Internet of Things (IoT) exponentially expands the web of devices connected to the Internet and allows them to talk to each other. Automated handling of electronic devices existed long before IoT and smart devices came to life, but IoT enables smart functions like voice enabled control and devices talking to each other (e.g., Alexa, Siri, Google Home) using low-cost wireless protocols including Wi-Fi and Bluetooth, and now ZigBee and Z-Wave protocols.

ZigBee and Z-Wave protocols are ideal for smart devices and smart home architecture as they are optimised for low data transmission and low power consumption as compared to Bluetooth or Wi-Fi protocols. Also, they can be transmitted in closed areas like smart homes covered with walls and furniture.

In a smart home architecture, smart devices are connected with a wireless network through a central hub which can be controlled by mobiles and tablets. Azure IoT Hub is such a gateway transmission service for connected devices in a cloud-based system.

CAD or computer aided design was widely used earlier for product development. Now, with digital architecture and device management in use, digital twins help to create dynamic, flexible, configurable and real-time extensible simulators. With IoT-based application designs, they play

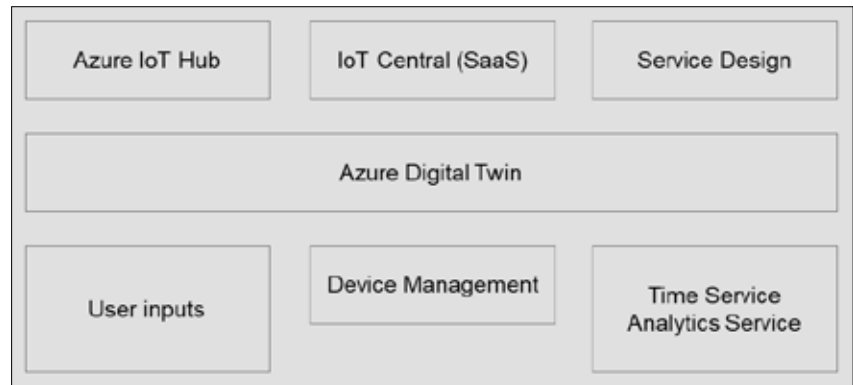


Figure 3: Azure Digital Twin components with IoT Hub

a vital role in product simulation, testing and inspection.

Microsoft Azure has introduced a sophisticated Digital Twin platform for such simulated modelling, based on a flexible open modelling language called Digital Twins Definition Language (DTDL). Its construct is the JSON-LD standard, and it can be tailored to any use case as needed.

Azure Digital Twins is more than a product simulator and can be used for service or process simulation. It utilises IoT reference architecture including Azure IoT Hub and other Azure functions and REST API services from the Azure platform. Azure Digital Twins can flexibly integrate with time series insights, storage, data lake services, or analytics services to prepare a complete (unified) simulated service or environment for product engineering services.

A lot of features are being included in Azure Digital Twins to build service simulation on-the-fly by combining other powerful features of the Azure platform.

Industrial use cases of Azure IoT services

▪ Predictive maintenance

Real-time monitoring: Azure IoT enables continuous monitoring of equipment health through sensors and IoT devices. This allows for the collection of data such as temperature,

vibration, and pressure.

Failure prediction: By applying machine learning algorithms, companies can predict equipment failures before they happen, reducing downtime and maintenance costs.

Cost efficiency: Proactive maintenance scheduling helps in optimising resource allocation and extending the lifespan of machinery.

▪ Supply chain optimisation

Inventory management: IoT devices provide real-time insights into inventory levels, helping businesses manage stock efficiently and reduce excess inventory.

Logistics tracking: Azure IoT facilitates the tracking of shipments, ensuring timely deliveries and improved logistics management.

Demand forecasting: By analysing data trends, companies can forecast demand more accurately, optimising production schedules and reducing waste.

▪ Energy management

Smart grids: Azure IoT supports the development of smart grids that adjust energy distribution based on real-time demand, improving efficiency and reducing costs.

Consumption analysis: Industries can monitor energy consumption patterns and identify areas for improvement, leading to significant cost savings.